PRADNYESH DIWAN

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Summary

Electronics & Communication Engineering undergraduate with hands-on experience in VLSI circuit design and embedded systems. Designed and simulated CMOS logic circuits in Cadence Virtuoso (gpdk90nm) and built ESP32-based automation systems. Experienced in executing electronics projects and skilled in team collaboration, leadership, and event management through active involvement in the VIT Dance Club.

Education

Vellore Institute of Technology Jan 2023 – Present
B.Tech, Electronics & Communication Engineering; CGPA: 8.94 *Vellore, India*

Anglo Urdu Boys High School Feb 2023
12th Grade (State Board) – 79.2% *Pune, India*

St. Joseph Convent School Aug 2021
10th Grade (CBSE) – 95% *Maharashtra, India*

Technical Skills

- **VLSI Design:** Cadence Virtuoso (Schematic, ADE-L, Symbol View), CMOS Logic Design, gpdk90nm Library, Transient Analysis
- **Programming/HDL:** Verilog, C, C++, Embedded C
- **Tools:** MATLAB/Simulink, Multisim, ESP32, Sensor Interfacing (Ultrasonic, LDR, DHT11)

Experience

Embedded System Design Intern – Maven Silicon May 2025

- Completed hands-on training in embedded systems architecture, hardware selection, firmware coding, and testing across multiple lab-based mini-projects.
- Built a real-time automation system on ESP32 integrating 3 sensors (ultrasonic, LDR, DHT11) for autonomous fan and light control.
- Debugged and validated embedded firmware through iterative test cycles, reinforcing hardware-software integration skills.

Projects

4-Bit Priority Encoder | *Cadence Virtuoso, gpdk90nm, ADE-L* 2025

- Engineered NAND2 and NAND4 logic gates at the transistor level (parallel PMOS pull-up, series NMOS pull-down networks) using the gpdk90nm CMOS technology library in Cadence Virtuoso.
- Performed transient simulation and waveform analysis in the ADE-L environment using VPULSE input stimuli, achieving 100% correlation between simulated outputs and the expected truth table across all input conditions.

ESP32-Based Smart Home Automation System | *ESP32, DHT11, Ultrasonic Sensor, LDR, 2-Channel Relay Module, Embedded C/C++* 2025

- Architected a dual-condition decision system on ESP32 that integrates ultrasonic (presence), LDR (ambient light), and DHT11 (temperature) sensor data to independently control fan and light relays..
- Implemented modular Embedded C firmware with dedicated functions for sensor acquisition, threshold-based logic, and relay actuation, enabling real-time processing and easy maintainability.
- Achieved sub-2-second response time from sensor input to relay activation, validated through structured unit, integration, and 6-hour continuous stress testing across all input combinations.

Verilog-Based Adaptive Traffic Light Controller | *Verilog HDL, Multisim* 2025

- Designed a sensor-driven traffic controller in Verilog that dynamically grants right-of-way to farm vehicles while preserving highway priority via an always-green default state.
- Modeled multi-state finite state machine (FSM) control logic to arbitrate competing traffic priorities under varying sensor input conditions.

Leadership & Extracurricular

2CC Coordinator – VIT Dance Club Present

- Planned and executed multiple large-scale dance events end-to-end, coordinating across diverse stakeholder groups to ensure on-time, smooth delivery.
- Strengthened leadership and cross-team communication by managing logistics and creative direction simultaneously.

- **Scored 97.3 percentile in JEE Mains; Qualified for JEE Advanced.**
- **Fluent in English, Hindi, and Marathi.**